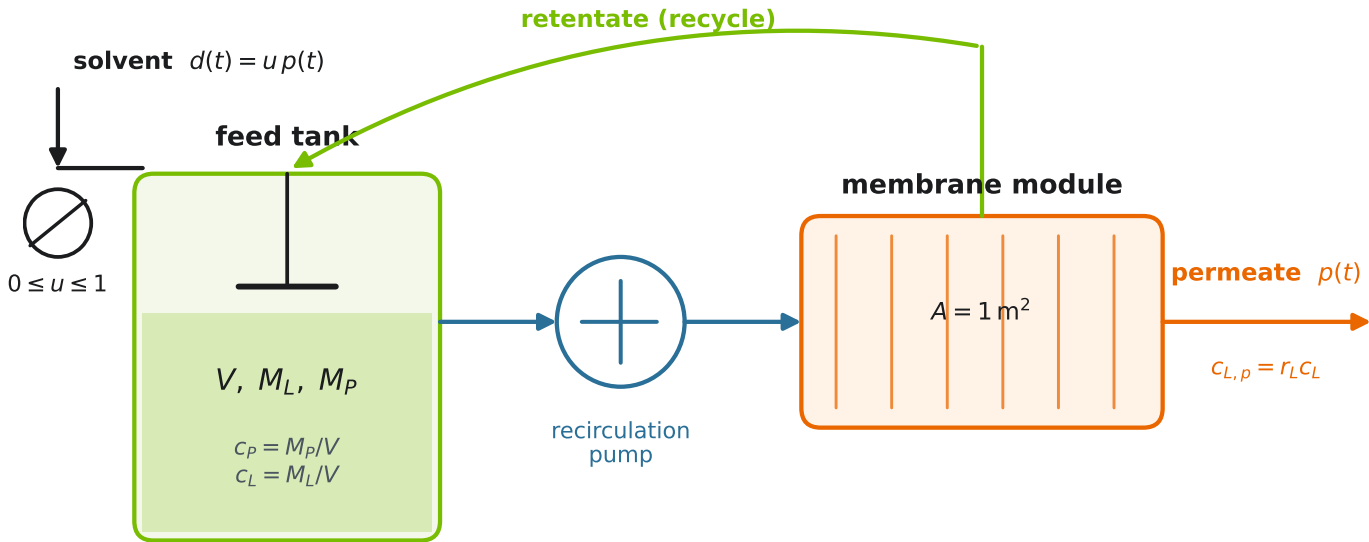




$$p = kA \ln(c_g/c_P) \quad \dot{V} = (u - 1)p \quad \dot{M}_L = -r_L C_L p \quad \dot{M}_P = -r_P C_P p$$

$$r_{\cdot} = \gamma / [1 + (\gamma - 1)e^{p/(k_M A)}], \quad \gamma = \alpha \text{ (lactose) or } \beta \text{ (protein)}$$